



Pre installation

Installation

Operating

**LC 20MT/O Chilled Water System for
GE Signa TwinSpeed MRI Equipment**

**Technical Publication 470
Revision 2**

The People of Ellis and Watts are committed to set goals and objectives to meet the expectations and needs of our customers. Responsibility and ownership at the point of operation will enhance the success of our mission

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REVISION HISTORY PAGE

REVISION NUMBER	DATE	REASON FOR CHANGE	AUTHOR
PRELIMINARY	05/23/01	NEW PRODUCT RELEASE	C.ROUNDTREE
A	07/01/01	REVISIONS PER GE	C. ROUNDTREE
B	08/06/01	REVISIONS PER GE	C. ROUNDTREE
C	08/31/01	REVISIONS PER GE	C. ROUNDTREE
0	09/25/01	INITIAL RELEASE	C. ROUNDTREE
1	10/19/01	UPDATE TO PAGES 17, 29, 46	C. ROUNDTREE
2	11/26/01	UPDATES TO PAGES 11, 15-20, 22-25, 44-46	C. ROUNDTREE

1 INTRODUCTION

1-1 CONTACT ELLIS & WATTS

If a non-service issue arises concerning the equipment or the documentation provided with the chiller you can contact Ellis & Watts in writing at the following address, by phone, or via E-mail at the Ellis & Watts web page.

**Ellis & Watts International, Inc.
Thermal Sales
4400 Glen Willow Lake Lane
Batavia, OH 45103**

Phone: 513-752-9000

Internet: WWW.ELLISWATTS.COM

Ellis & Watts provides 24-hour service support in the event that the Ellis and Watts equipment fails to perform properly. Service calls are processed more easily if you provide the service support department with the chiller **model number, serial number, the site name and location, and a description of the problem including the alarm or fault displayed on the operator interface**. The service support department will dispatch a qualified local service technician to correct the problem.

The 24 hour service support department phone numbers are:

**Domestic Toll Free: 1-888-744-3195
International: 513-768-3195**

1-2 WARRANTY REPLACEMENT PART PROCEDURE

To initiate a warranty claim with Ellis & Watts, contact the Thermal Sales Department. The equipment model number and serial number are required to process a warranty claim. The Thermal Sales representative will issue a Return Goods Authorization (RGA) number. The RGA number must be on the returned part's package.

NOTICE

Items returned without an RGA number will not be processed until the necessary information is obtained from the sender.

1-3 INSTALLATION TOOLS & EQUIPMENT

Item Description	QTY	Used For	Supplied By
Crowbar, 2 foot (.61m)	1	Uncrating Equipment	Installer
Pry bar, lever, 5000 lb. (2250kg)	1	Lifting Equipment	Installer
4 Conductor (4AWG/25mm ²) power cable minimum	1	Power for TSCC Cabinet	Installer
4X4X4 inch wood block (10.16X10.16X10.16 cm)	4	Supporting TSCC Cabinet, Hose Assembling	Installer
Wood Block Approximately 1X4X6 inch (2.54 x 10.16 x 15.24cm)	1	Supporting TSCC Cabinet	
Metal Snips	1	Cut banding Straps	Installer
2 Eye, Lift Straps 2 inch W x 18ft. Lg (5.08cm W x 5.49m Lg)	2	Lifting TSCC off Shipping Skid	Installer
C Clamp, 10 inch (25.4cm)	2	Securing Straps	Installer
Drill 3/8 inch minimum	1	Drill mounting holes	Installer
3/8 inch (M10) Hardware Length to be determined	4	For tie down bracket	Installer
3/4" Copper hard Pipe	200 feet maximum	For Cryo Cooler Connection	Installer
3/4" Ball Valves	2	For Cryo Cooler Connection	Installer
1/2" Hose Barbs	4	For Cryo Cooler Connection	Installer
1" ID Water Hose	200 feet maximum	For Gradient Coil Connection	Installer
Light oil or soapy water	1 oz	Lubricating Hose Barbs	Installer
LocTite 567 or equivalent	4 oz	Sealing Pipe Threads	Installer
Razor Knife	1	Cutting Hose	Installer
10 foot (3.05 m) Stepladder	2	Uncrating Equipment	Installer
Crane (Installer to determine size and reach)	1	For Rooftop or Wall Mount	Installer
Forklift, 2 ton (1814 kg) Minimum	1	Lifting and Moving TSCC and Associated Equipment	Installer

Table 1 **INSTALLATION TOOLS AND SUPPLIES**

1-3-1 Installation Responsibility Matrix

	Cust. Contr.	GE Service	E&W Service	Who Pays
Once Cooling Cabinets Arrive				
Unload TSCC from Truck	X			Customer
Install concrete pad	X			Customer
Move TSCC to outdoor location and mount. (Use seismic Tie down brackets where required by law). See Section 3-5, 6	X			Customer
Connect customer supplied power cable from MDP to TSCC.	X			Customer
Install Remote Control Panel in Equipment Room.	X			Customer
Install interconnecting cables between Remote Control Panel and TSCC. See Section 5-1	X			Customer
Install copper water lines with shut off valves from TSCC to Equipment Room.		X		GE
Once MR System Arrives				
Install water lines from Equipment Room to Gradient Coil. See Section 5-1		X		GE
Fill TSCC with Propylene Glycol		X		GE
Start up TSCC and ensure proper operation		X		GE
Verify proper operation per E&W Chiller Startup Check List after Cryo Cooler Compressor load is present			X	E&W

Table 2 **INSTALLATION RESPONSIBILITY MATRIX**

1-4 EXPLANATION OF NOTES, CAUTIONS, AND WARNINGS

A **NOTE** is used to provide supplemental information or direction.



A **CAUTION** is used to call particular attention to a step of a procedure, which, if not strictly followed, could result in damage to, or destruction of equipment.



A **WARNING** is used to call attention to a particular step of a procedure, which, if not strictly followed, could result in serious injury, or death of personnel.

1-5 SAFETY WARNINGS

WARNING

NO PERSON SHALL REACH INTO THE UNIT ENCLOSURE, WHICH CONTAINS LIVE ELECTRICAL CIRCUITS AS WELL AS ROTATING PARTS, FOR THE PURPOSE OF INSPECTING, SERVICING, ADJUSTING, OR MAINTENANCE EXCEPT IN THE PRESENCE OF SOMEONE CAPABLE OF RENDERING AID. CONTACTING SUCH ITEMS COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

PERSONNEL WORKING NEAR HIGH VOLTAGES SHOULD BE IN THE PRESENCE OF SOMEONE HAVING RESUSCITATION SKILLS. COMING IN CONTACT WITH HIGH VOLTAGE COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

TURN OFF AND TAG OUT THE MAIN POWER AT THE DISCONNECT, DURING ANY SERVICING OR MAINTENANCE OPERATIONS. REMOTE STARTING COMPONENTS OR DIRECT CONTACT WITH ELECTRICAL COMPONENTS MAY RESULT IN INJURY TO PERSONNEL.

WARNING

TURN OFF AND TAG OUT THE DISCONNECT TO THE UNIT BEFORE WORKING ON CONTACTORS OR OTHER COMPONENTS IN THE ELECTRIC BOX. WHEN THE MAIN POWER SUPPLY TO THE UNIT IS ON, THE LIVE SIDE OF EACH CONTACTOR CONTAINS HIGH VOLTAGE REGARDLESS OF ANY OTHER SWITCH SETTING. DO NOT ATTEMPT MAINTENANCE PROCEDURES INSIDE EQUIPMENT WITH HIGH VOLTAGE ON. REMOVE POWER AND TAG OUT THE DISCONNECT. GROUND ELECTRICAL COMPONENTS PRIOR TO TOUCHING TO PREVENT BODILY INJURY.

SAFETY WARNINGS CONTINUED

WARNING

NO SMOKING OR OPEN FLAMES WITHIN 50 feet (15 M) OF REFRIGERANT SYSTEM. REFRIGERANT COMING IN CONTACT WITH LIT CIGARETTES CAN CREATE AN IRRITATING AND POSSIBLY TOXIC GAS.

WARNING

STAND CLEAR DURING LIFTING OPERATIONS. OPERATOR ERROR AND/OR EQUIPMENT MALFUNCTION CAN CREATE OPPORTUNITIES FOR PERSONAL INJURY.

WARNING

THE UNIT HAS A CLOSED LOOP HIGH PRESSURE REFRIGERANT SYSTEM. DO NOT CLOSE, OPEN OR ADJUST VALVES IN THE REFRIGERANT SYSTEM UNLESS A SPECIFIC INSTRUCTION STATES THAT VALVES MUST BE OPENED, CLOSED OR ADJUSTED. PERSONNEL COULD BE INJURED OR EQUIPMENT COULD BE DAMAGED.

WARNING

BEFORE BRAZING OR SOLDERING ANY REFRIGERATION COMPONENT, RECOVER REFRIGERANT PER US ENVIRONMENTAL PROTECTION AGENCY (EPA) APPROVED PROCEDURE TO RELEASE THE PRESSURE FROM THE REFRIGERATION SYSTEM. SUDDEN DECOMPRESSION OF REFRIGERANT PIPING CAN CAUSE SERIOUS PERSONAL INJURY.

WARNING

USE EXTREME CARE WHEN BRAZING OR SOLDERING IN THE PRESENCE OF R-134A. WHEN R-134A COMES IN CONTACT WITH HOT METAL, AN IRRITATING GAS CAN FORM. HAVE SUFFICIENT FRESH AIR AND VENTILATION WHEN BRAZING, SOLDERING OR USING A HALIDE TORCH LEAK DETECTOR.

SAFETY WARNINGS CONTINUED

WARNING

THE COMPRESSOR MUST ALWAYS BE OPERATED WITH THE COMPRESSOR DISCHARGE SERVICE VALVE IN THE BACK-SEAT (FULLY OPEN) POSITION. FAILURE TO DO SO CAN CAUSE SERIOUS COMPONENT FAILURE.

WARNING

WHEN AN ABNORMAL CONDITION IS NOTED AND PROCEDURES IN THIS MANUAL DO NOT SPECIFICALLY RELATE TO THE NOTED CONDITION, WORK SHALL BE STOPPED. TECHNICALLY QUALIFIED PERSONNEL SHALL GIVE GUIDANCE ON THE ABNORMAL CONDITION BEFORE WORK CAN BE CONTINUED, TO PREVENT POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE.

2 SHIPPING AND DELIVERY DATA

The General Electric Medical Systems GEMS part number 2281346-2 High Voltage Air Cooled Unit (380V – 50 Hz, 480V – 60 Hz) is the Ellis and Watts chiller model LC20MT/O and is designated as the TSCC (Twin System Cooling Cabinet).

2-1 SHIPMENT

For domestic shipment, the TSCC will be shipped to the construction site. International shipment shall be shipped to a collection point where all components for the magnet will be collected and shipped to the construction site.

Shipping Dimensions

Unit	Height	Width	Depth	Weights
TSCC	77 inches (196 cm)	50 inches (127 cm)	80 inches (203.2 cm)	1320 lbs. (598.7 kg)
Ship Loose Box		28.5 inches (72.5 cm)	58.8 inches (149.4 cm)	250 lbs (113.4 kg)

Table 3 **SHIPPING DIMENSIONS**

The ship loose items are boxed and shipped together on a wooden skid. Domestic shipments are covered with cardboard and banded on wooden skids. International shipments are domestic packages shipped in crates.

2-2 STORAGE REQUIREMENTS

The non-operation environmental conditions are as follows:

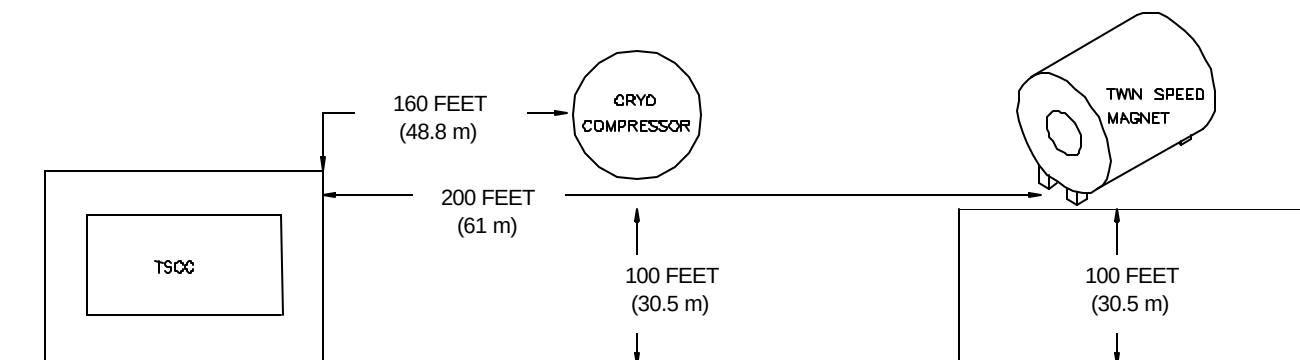
ITEM	REQUIREMENT
Ambient Temperature	-40°C to 70°C
Humidity	5% to 100% RH
Altitude	984 feet (300 m) below sea level 17,060 feet (5,200 m) above sea level
Acceleration	+/- 1.5 G any axis
Magnetic Field	30 gauss
NOTE: Do not expose to water (rain, snow)	

Table 4 **ENVIRONMENTAL CONDITIONS**

NOTE The TSCC should be kept in its shipping crate until it has been moved to the installation site. When uncrating, care should be taken to prevent damage.

3 PRE-INSTALLATION REQUIREMENTS

3-1 EQUIPMENT PLACEMENT



The maximum distance from the TSCC to the magnet is 100 feet (30.5 m) vertical and 200 feet (61 m) total. The maximum distance from the TSCC to the Cryo Cooler is 100 feet (30.5 m) vertical and 160 feet (48.8 m) total.

Illustration 1 **PLACEMENT OF TSCC TO MAGNET**

3-2 SITE ENVIRONMENT

3-2-1 Operating Specifications

ITEM	REQUIREMENT
Ambient Temperature	-30° C to 50° C
Humidity	5% to 100%
Altitude	984 feet (300 m) below sea level to 11,800 feet (3,597 m) above sea level
Noise Levels	75 dBa AT 1 meter in all directions
Magnetic Field	30 Gauss

Table 5 **OPERATING SPECIFICATIONS**

3-2-2 Power Requirements

ITEM	REQUIREMENT
Wire NOTE 1	6 AWG (minimum) 3 AWG (maximum) 3 conductor w/ground
Power Consumption Continuous NOTE 2	18.3 kVA (maximum)
Voltage Requirements	400 VAC $\pm$ 10% @ 50 Hz $\pm$ 3 Hz 480 VAC $\pm$ 10% @ 60 Hz $\pm$ 3 Hz
Circuit Size	50 AMP (maximum)
Digital Interconnect Cable NOTE 3	Cable Length – 100 feet (30.5 m) Cable Diameter - .625 inches (1.59 cm) Connector Diameter – 1.97 inches (5.00 cm)
Analog Interconnect Cable NOTE 3	Cable Length – 100 feet (30.5 m) Cable Diameter - .745 inches (1.89 cm) Connector Diameter – 1.34 inches (3.40 cm)
NOTE: 1The chiller will be hardwired to the MDP with a four conductor (6 AWG minimum/3 AWG maximum) power cable supplied by the installer. NOTE 2 Heat load of RCP is 10 Watts NOTE 3: If cables are run in conduit, consideration must be made for the connector diameter.	

Table 6 **POWER REQUIREMENTS**

3-3 EQUIPMENT CLEARANCE REQUIREMENTS

It is necessary to maintain proper distances at the top, front, back and sides of the TSCC cabinet in order to have a good service area.

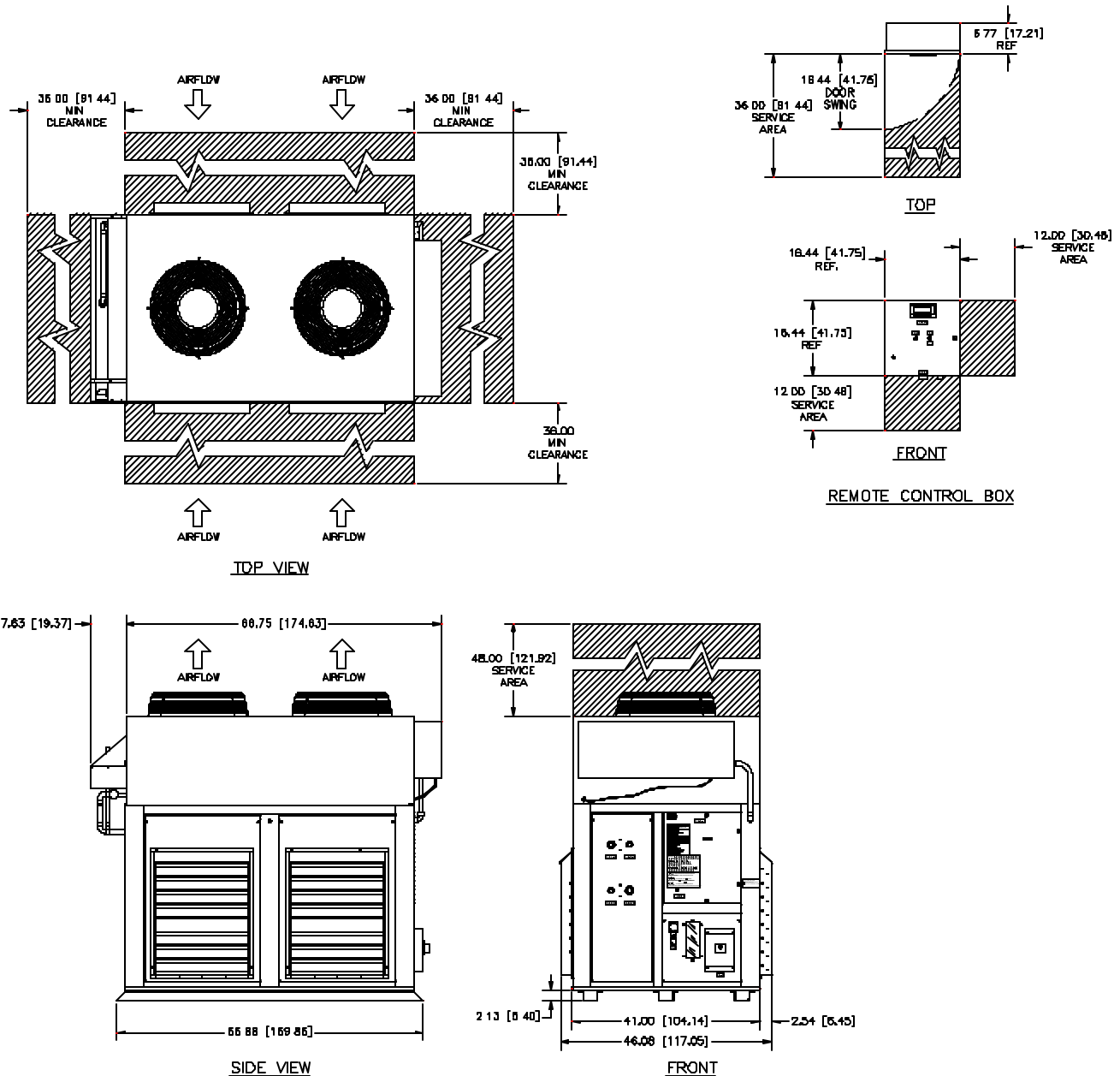
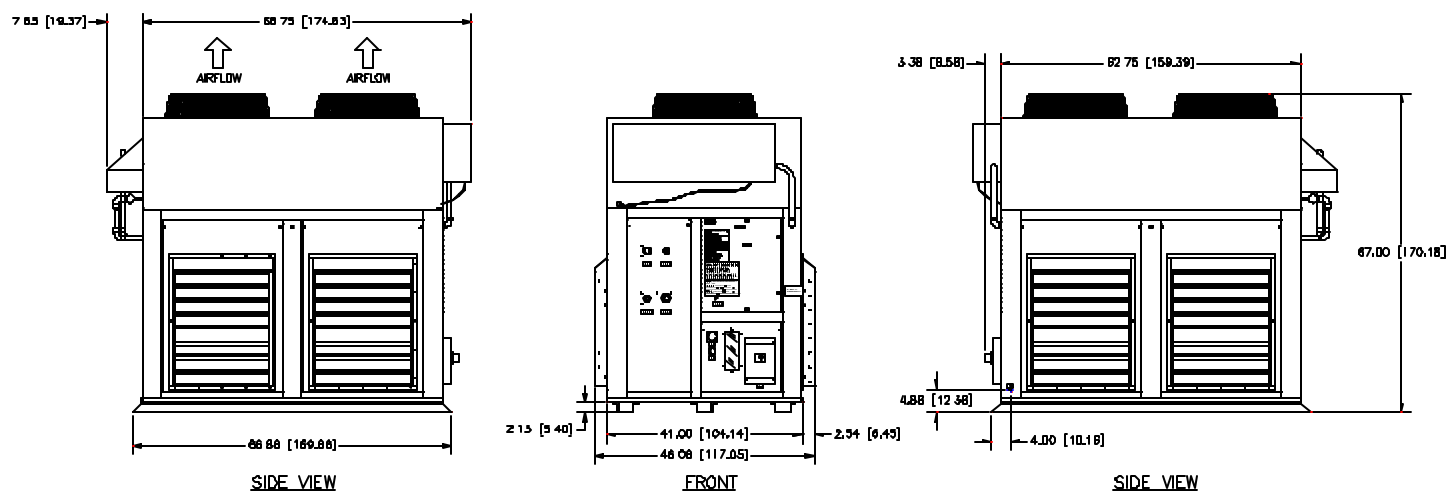


Illustration 2 **NECESSARY CLEARANCES FOR TSCC** ([] = cm)

3-4 COMPONENT DIMENSIONS**Uncrated Dimensions**

Component	Height	Width	Depth	Installed Weight
RCP	16.44 inches (41.76 cm)	16.44 inches (41.76 cm)	6.77 inches (17.2 cm)	30 lbs. (13.6 kg)
TCC	66.88 inches (169.88 cm)	45.5 inches (115.57 cm)	74.83 inches (190.10 cm)	1200 lbs (544 kg)

Table 7 **UNCRATED DIMENSIONS****Illustration 3 TSCC OVERALL DIMENSIONS ([] = cm)**

The clearance from the bottom of the TSCC cabinet to the floor is approximately 2 inches (5 cm) therefore doorway threshold should be as smooth as possible or special considerations have to be made to move cabinet over a threshold height of 1.5 inches (3.81 cm).

3-5 FLOOR LOADING AND WEIGHTS

EQUIPMENT	INSTALLED WEIGHT
TSCC	1200 lbs. (544 Kg)

Table 8 FLOOR LOADING AND WEIGHTS

3-6 ANCHORING AND SEISMIC CONSIDERATIONS AND CONCRETE PAD

The TSCC cabinet has pre-drilled holes that should be used for mounting, which should be used to aid in preventing the cabinets from shifting during a seismic event.

Sites requiring seismic anchoring by code should have the site architect and engineer review the response spectra and/or United Builders Code (UBC) for their location and then contact Ellis and Watts International (Call 513-752-9000) for assistance in seismic planning of the site.

The TSCC can be mounted on a roof or concrete slab (ground level installation). Any other mounting configurations should be discussed with the Ellis & Watts engineering department. A concrete pad used for mounting the TSCC unit should be installed level and be properly supported to prevent settling. A one piece slab 70.00 inches (177.80 cm) x 120.00 inches (304.80 cm) of 2500 psi (17.23 Mpa) concrete a minimum of 4 inches (10.16 cm) deep is recommended. The concrete footing should meet or exceed the local code requirements.

NOTES:

- 1 MINIMUM 4" DEEP CONCRETE PAD OF 2500 PSI CONCRETE. ADDITIONAL CONCRETE DEPTH MAY BE REQUIRED AS DICTATED BY LOCAL CODES.

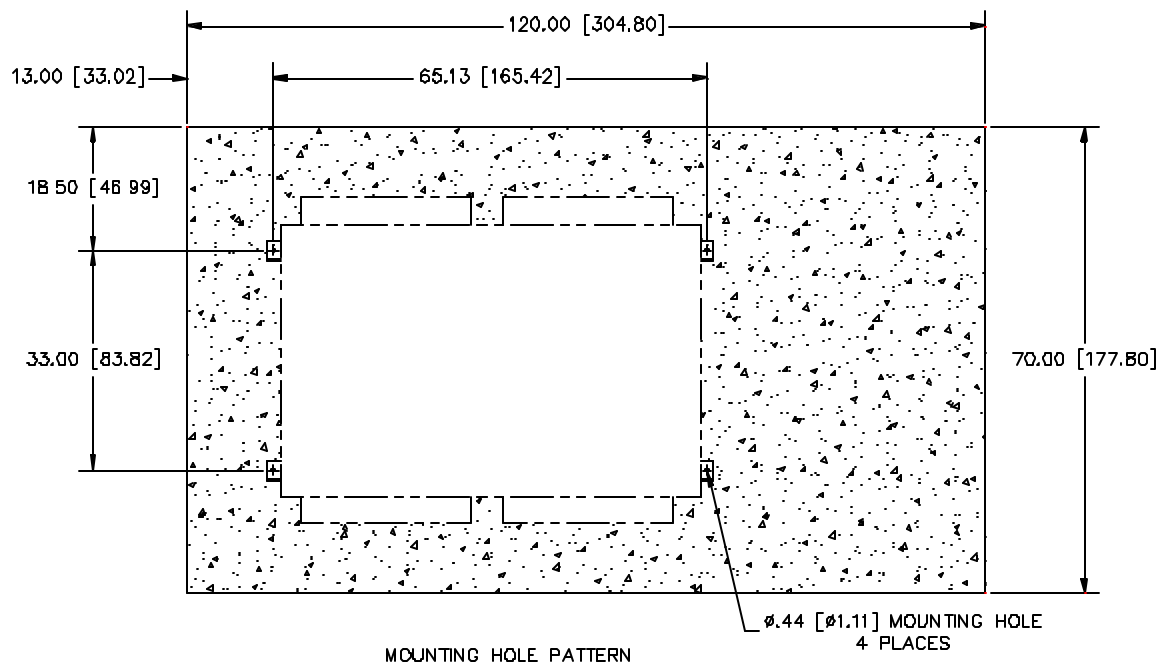


Illustration 4 TSCC MOUNTING PAD DIMENSIONS AND PATTERN ([] = cm)

4 UNCRATING AND INSTALLATION PROCEDURES

4-1 UNCRATING AND RECEIVING INSPECTION

NOTE Steps 1 through 3 are for International shipments only.

4-1-1 TSCC

1. Using a #2 Phillips screwdriver, unscrew the wood screws holding on the top of the crate.

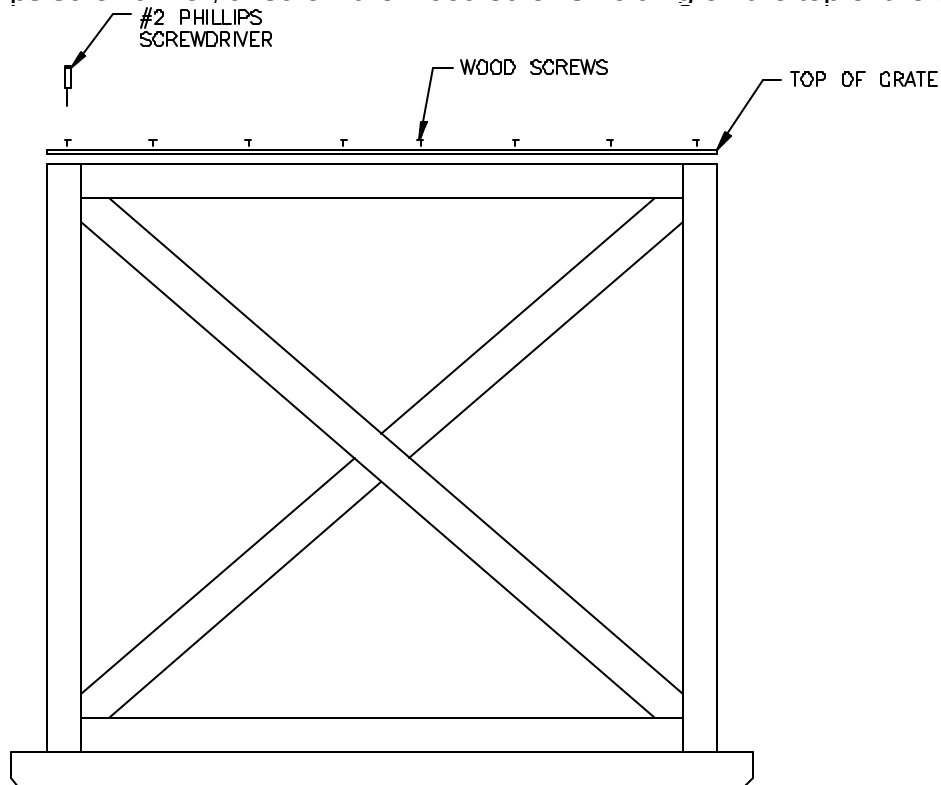
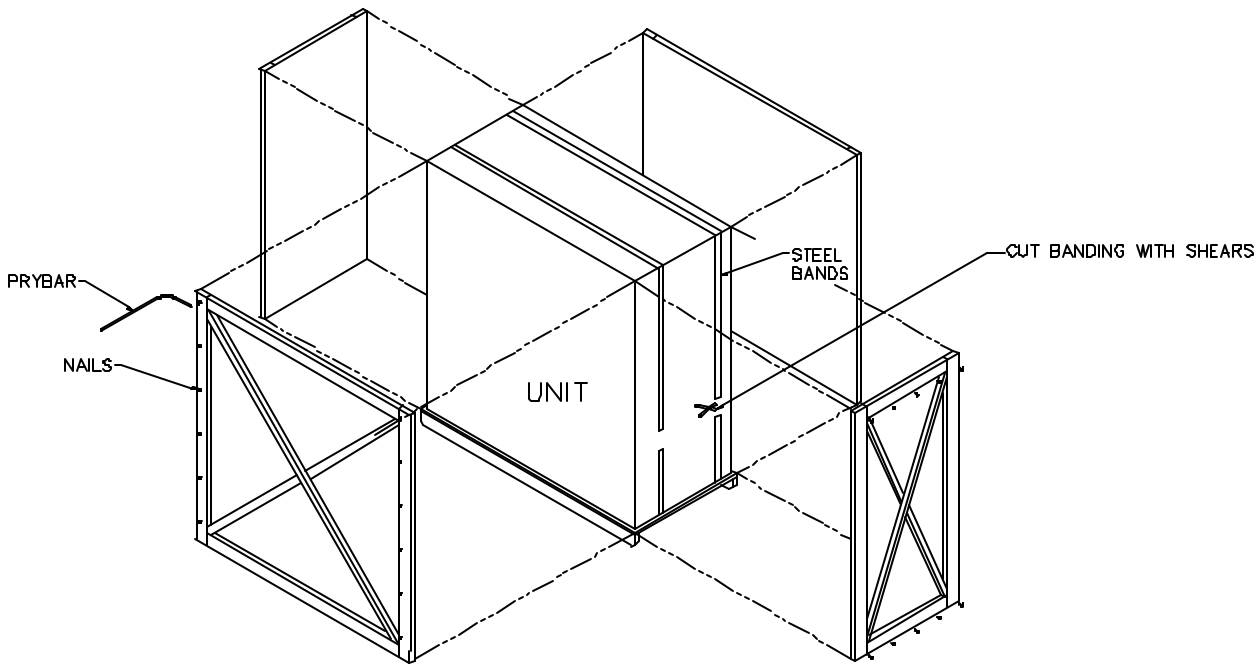


Illustration 5 INTERNATIONAL CRATE EXAMPLE

2. Remove top of the crate and place in an area clear of the uncrating area.
3. Using a pry bar carefully remove the front, sides, and back of the crate and place in an area clear of the uncrating area.

Illustration 6 **UNCRATING****CAUTION**

Banding straps can be under high tension and can whip out when cut.

4. Remove banding by cutting with metal cutting hand shears.
5. CAREFULLY remove plastic.
6. Inspect equipment for any damage. Contact Ellis and Watts and the shipping company as soon as possible and report any discovered damage.
7. There is a separate skid with a box on it that comes with the TSCC that contains ship loose items. These items are packaged inside the box and labeled accordingly. These items should be placed in a secure location for later use during installation. Boxes can be inventoried using the list in section 4.2.
8. Using a fork lift insert the forks between the board slats from front to back on the TSCC.

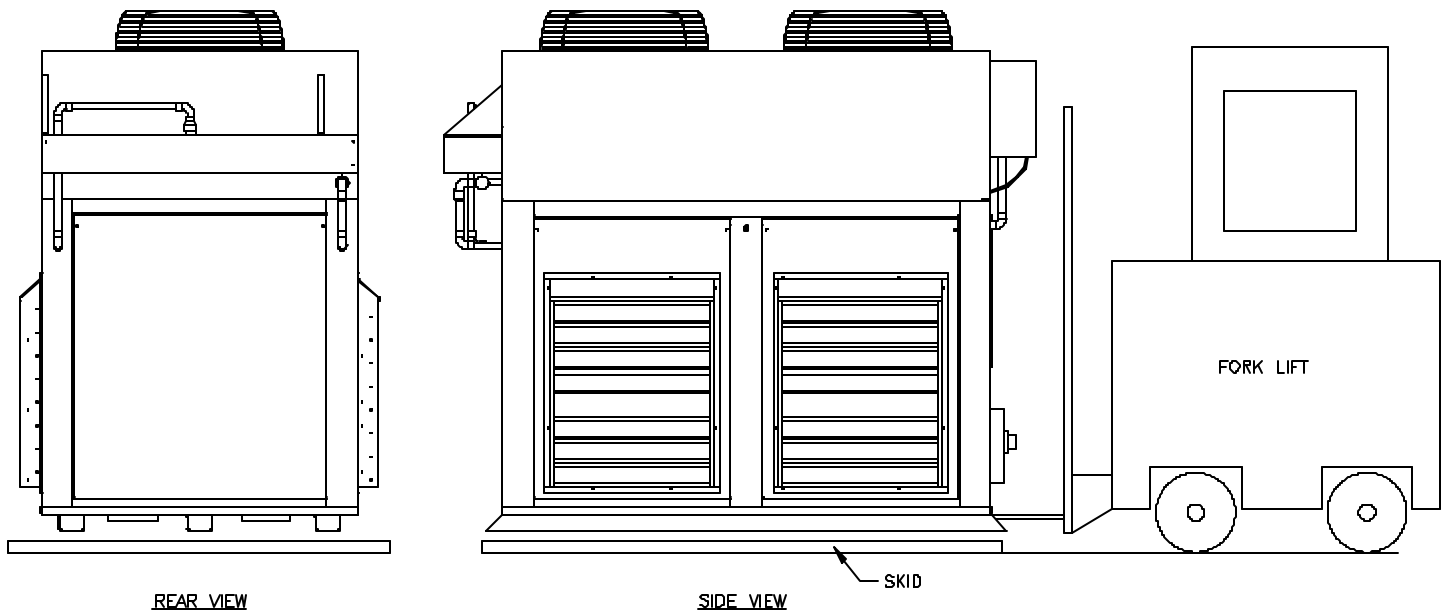


Illustration 7 **ACCEPTABLE LIFTING METHOD**

WARNING

STAND CLEAR DURING LIFTING OPERATIONS. OPERATOR ERROR AND/OR EQUIPMENT MALFUNCTION CAN CREATE OPORTUNITIES FOR PERSONAL INJURY.

12. Gently lift TSCC cabinet until clear of shipping skid.
13. Remove skid and carefully move unit to smooth level area.

The previous steps define the preferred method of removing the TSCC cabinet from the skid. Other methods of removing the TSCC cabinet from the shipping skid may be acceptable but could involve many more detailed steps. It is up to the judgment of the riggers and movers to determine the specific method of handling the TSCC cabinet.

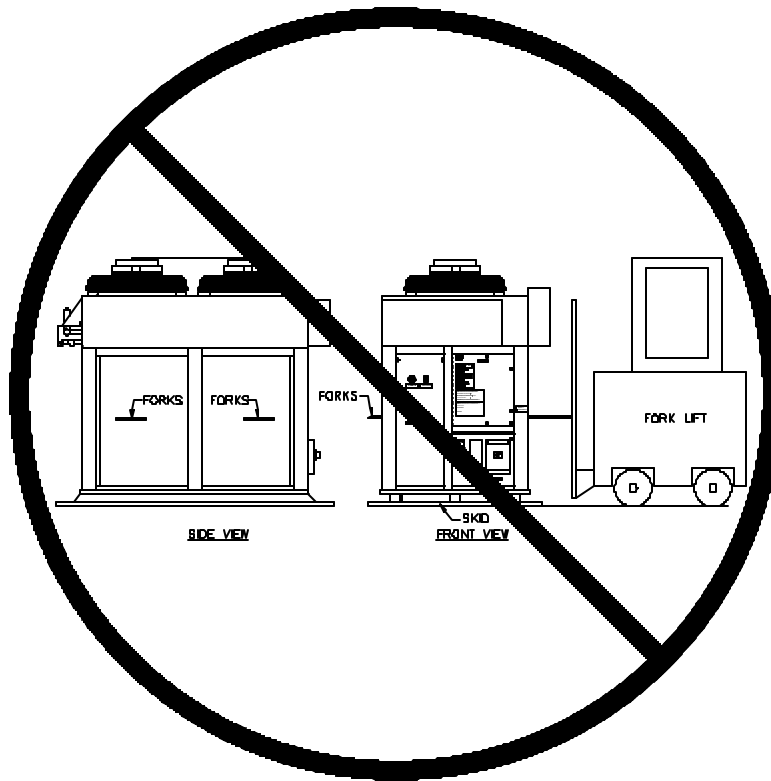


Illustration 8 **UNACCEPTABLE ALTERNATIVE LIFTING METHOD**

4-2 SHIP LOOSE ITEMS

The following tables list the ship loose items that are necessary for the proper installation of the Ellis and Watts equipment. These items were shipped with the TSCC. The following tables can be used to verify the inventory of the shipped loose items.

The TSCC is shipped with the following field installation kits.

No.	Part No.	Ship Loose Items Description	QTY EA
1	FI-0099	Field Installation Unit Mounting	1

Table 9 **INSTALLATION MOUNTING INSTALL KIT**

No.	Part No.	Ship Loose Items Description	QTY EA
1	J1538-1205	Cable Assembly, Digital, 100 FT. LG.	1
2	J1538-1204	Cable Assembly, Analog, 100 FT. LG.	1
3	FI-0100	Field Installation Electrical/Power/Comm	1

Table 10 **ELECTRICAL/POWER/COMM INSTALL KIT**

No.	Part No.	Ship Loose Items Description	QTY EA
1	SS1045-08	Tube, ½ inch Push-Lok	120 feet
2	62-1-189	Clamp, Hose, ½ inch	2
3	FI-0101	Field Installation Hose Connections	1

Table 11 **HOSE CONNECTIONS INSTALL KIT**

No.	Part No.	Ship Loose Items Description	QTY EA
1	T2625-279A	Filter, 10 inch Cartridge (Polyethylene)	1
2	SS627-7-A20	Fuse, Time Delay, 1.0 Amp, 500V	2
3	E&W P/N J1520-5050	Propylene Glycol GE P/N 229767Z	24 GAL
4	FI-0102	Field Installation Spare Parts Kits	1

Table 12 **SPARE PARTS KIT**

No.	Part No.	Ship Loose Items Description	QTY EA
1	FI-0107-01	Pump Purge Assembly	1
2	FI-0107	Field Installation Pump Purge Assembly	1

Table 13 **PUMP PURGE ASSEMBLY**

No.	Part No.	Ship Loose Items Description	QTY EA
1	J1539-0602	Remote Control Panel	1

Table 14 **REMOTE CONTROL PANEL**

5 EQUIPMENT INSTALLATION

5-1 TSCC

5-1-1 Electrical Installation

Power Wiring

Connecting power to the TSCC should only be performed by trained electrical installation personnel that have a good knowledge of lock-out/tag-out procedures. Short circuit and field wiring must be accomplished in accordance to the National Electrical code, ANSI/DEPA No. 70 (latest edition) and or the Canadian Electrical Code, Part 1, CSA Standard C22.1 (latest edition) and local laws and regulations as they exist.

Power Requirement

480 VAC $\pm$ 10% @60 Hz $\pm$ 3 Hz, 3 phase, WYE or Delta

or

400 VAC $\pm$ 10% @50 Hz $\pm$ 3 Hz, 3 Phase, WYE or Delta

Full Load current is 23.0 kVA.

Refer to drawing FI-0100 that comes with the ship loose parts, for detailed illustrations.

1. Locate the power disconnect switch on the lower right side of the TSCC.

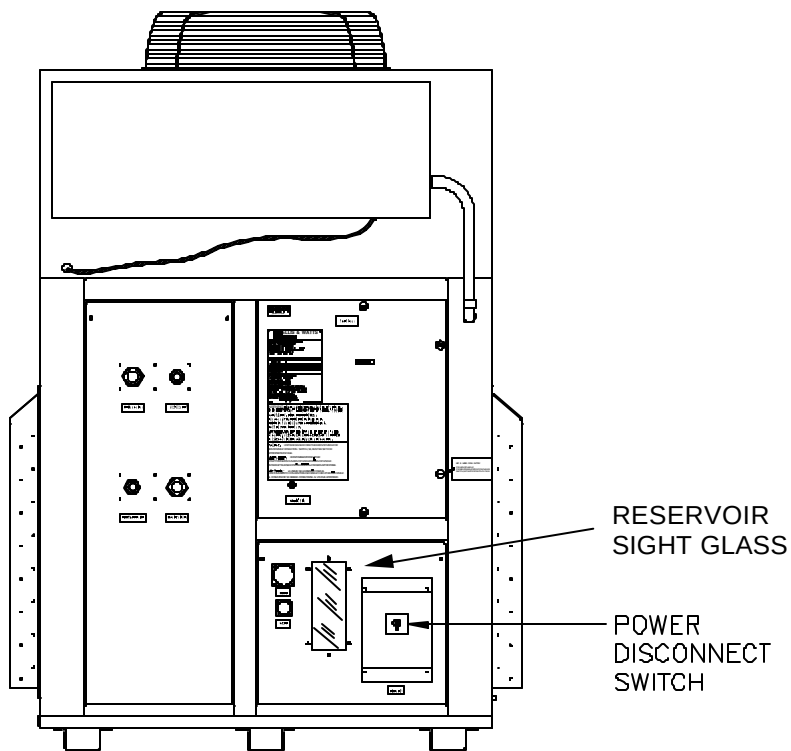


Illustration 9 JUNCTION BOX LOCATION

Ellis & Watts International, Inc.
Batavia, Ohio 45103

2. Remove the cover by loosening the four cover screws and sliding the cover off.

NOTE: Ellis & Watts recommends using a minimum conductor size of 6 AWG. Maximum size recommended is 3 AWG. Ellis & Watts recommends using 1 inch (minimum) Liquidtight (or equivalent) conduit and fittings. The installer supplies wire, conduit, and fittings.

3. Route the four power conductors through the 1 inch (2.54 cm) knockout provided on the junction box.

NOTE: It is very important to maintain correct phasing.

4. Strip the wire and insert into the terminals. When the conductor is properly installed, there should be no bare wire exposed. Pay special attention to the phase rotation.

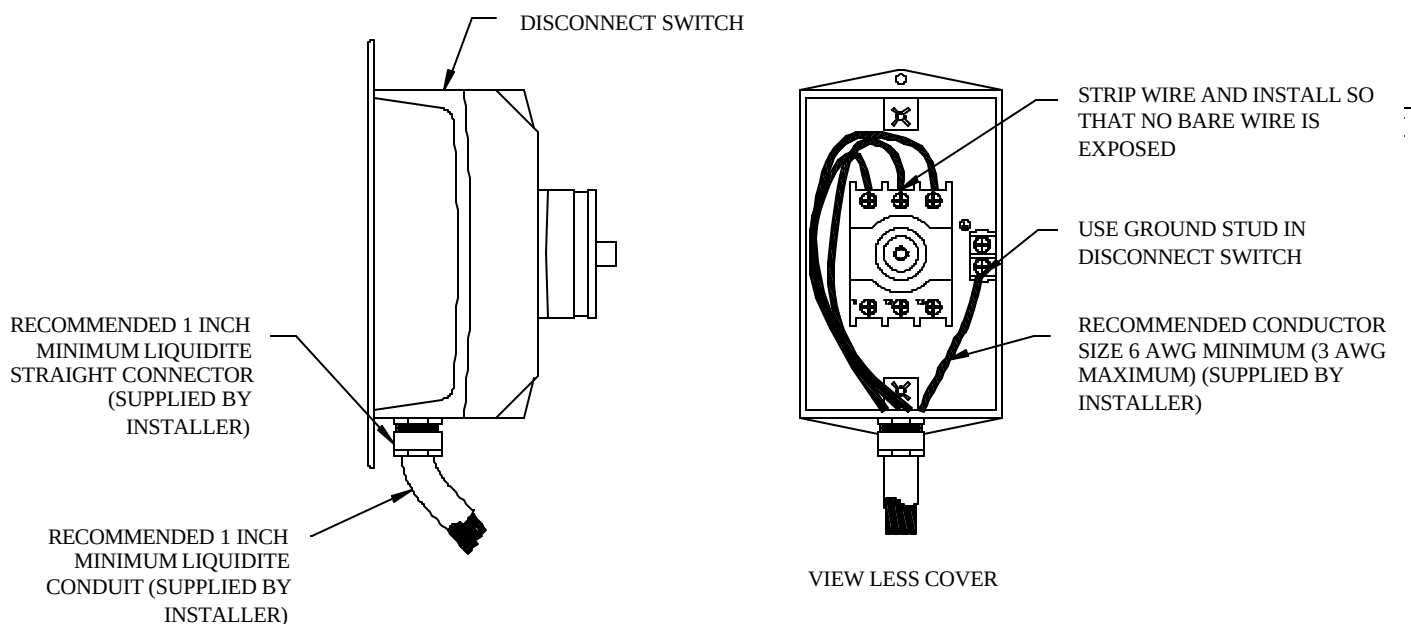


Illustration 10 JUNCTION BOX WIRING

5. Attach the ground wire to the ground stud provided on the junction box.
6. Reinstall the cover.

5-1-2 Assembling The Interconnecting Hoses

1. Verify the parts and quantities from the FI-0101 drawing. Refer to table 11.

Refer to drawing FI-0101 that comes with the ship loose parts, for detailed illustrations.

The following supplies are supplied by the installer and are required for the proper installation of the interconnecting hoses.

- Light oil or soapy water
 - Loctite 567 or equivalent
2. Cut 2 pieces of 1/2 inch tubing 60 feet (18.3 m) long cleanly and squarely with a razor knife.
 3. Lubricate the fitting and/or hose I.D. with a light oil or soapy water only. Do not use heavy oil or grease.
 4. Insert the fitting into the hose until the fitting is in the hose.

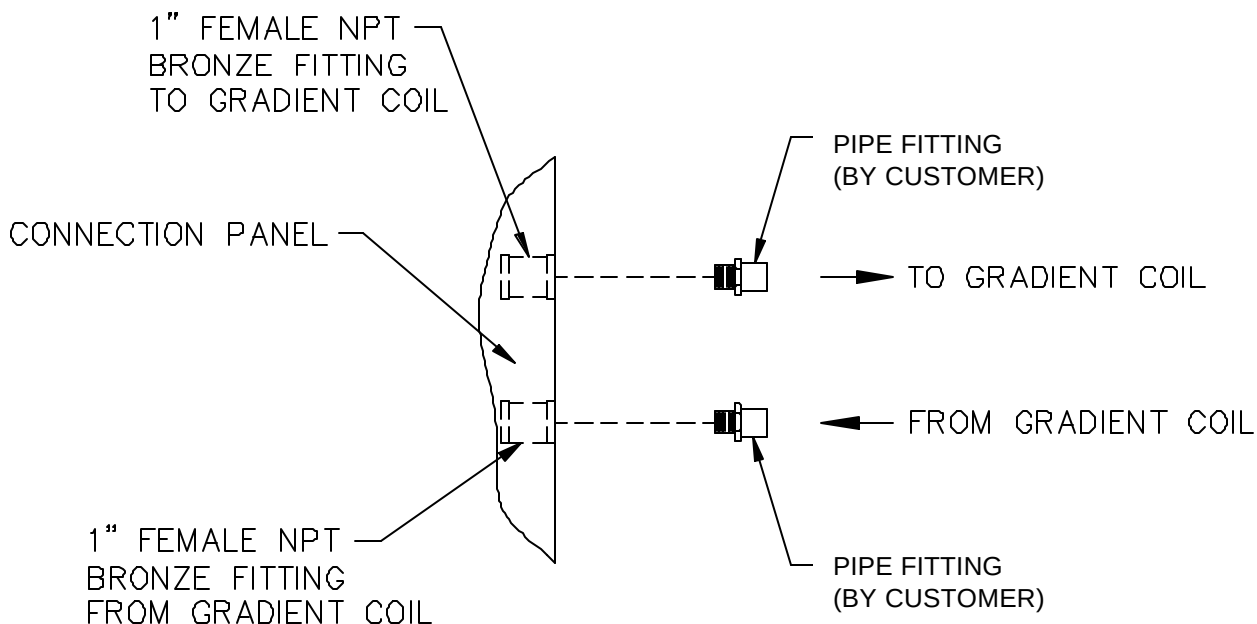
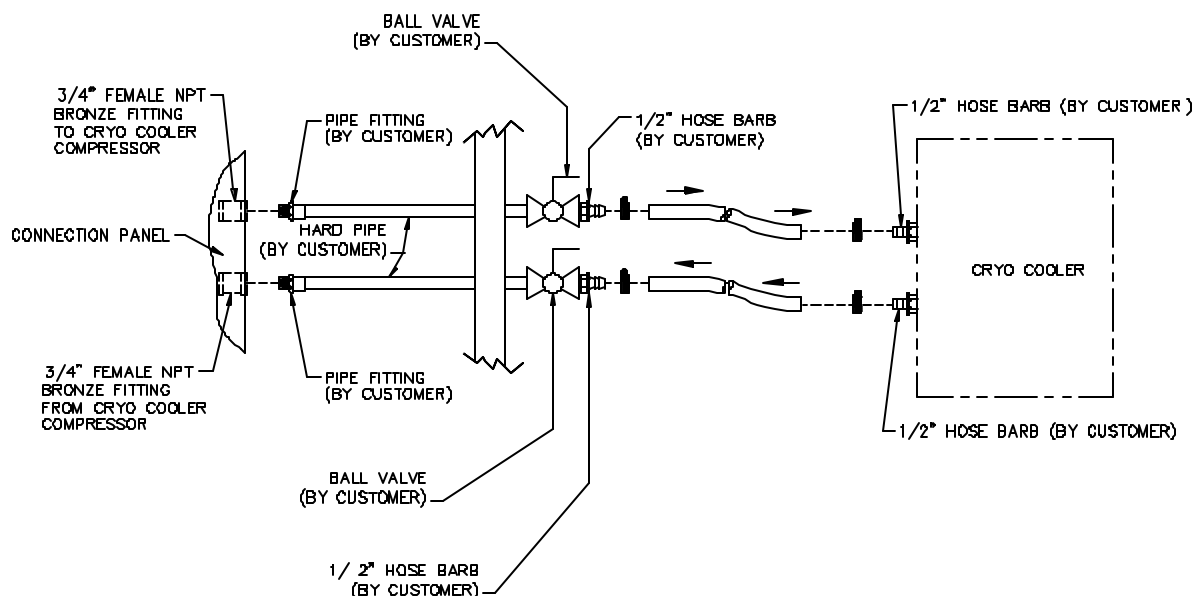


Illustration 11 **PIPE FITTING ASSEMBLY**

5. Place the end of the fitting against a wood block. Grip the hose firmly at approximately 1 inch (2.54 cm) from the end of the hose. Push with steady force until the end of the hose bottoms on the fitting and is covered by the yellow plastic cap.

Illustration 12 **TOTAL HOSE ASSEMBLY**

6. Apply a generous amount of LocTite 567 or equivalent to the male pipe threads of each pipe fitting.
7. Assemble and mark each hose per table.

HOSE SECTION Feet (m)	QUICK DISCONNECT COUPLER	MARK HOSE AS:
100 (30.5)	Male	Gradient coolant from TSCC
100 (30.5)	Female	Gradient coolant to TSCC
60 (18.3)	Male	Cryo cooler coolant from TSCC
60 (18.3)	Female	Cryo cooler coolant to TSCC

Table 15 **HOSE AND QUICK DISCONNECT**

8. TIGHTEN EACH THREADED CONNECTION SNUGLY TO PREVENT LEAKS.
9. Using hose clamps attach each hose assembly to appropriate connection of Cryo Cooler.

6 SYSTEM OPERATION

The following procedures are intended to give instruction on the TSCC system controls and routine operation. Initial start-ups after installation, will be performed by GE personnel and system operation will be verified after installation by an Ellis and Watts approved service provider. Start-ups after repairs should be conducted by the customers service provider. It is assumed at this point, that the TSCC and associated equipment have gone through initial startup previously and is ready for operation. If the unit has stopped due to a failure, refer to section 6-1-4.

6-1 OPERATING PROCEDURES

6-1-1 Starting Up The TSCC and Associated Equipment

1. Verify that all fluid lines are connected and not leaking. The fluid lines are located on the rear of the unit.
2. Check the coolant level in the reservoir. Verify that the coolant level is between the add and full labels. The reservoir sight glass is located on the front of the unit (See Illustration 9)



CAUTION

During periods where compressor crankcase heaters are without power, liquid refrigerant can accumulate in compressor crankcase. Restarting compressor without allowing sufficient time for crankcase heaters to vaporize liquid refrigerant can cause serious internal damage to compressor.

3. The Control Power Switch is located on the Remote Control Panel, (see Illustration 14). Verify toggle switch is switched off. If the main disconnect switch is OFF, set to ON by rotating red handle clockwise until handle is in vertical position. The main disconnect is located at the lower right front of the TSCC unit (see Illustration 9). **(Wait a minimum of two hours before continuing.)**
4. Verify that disconnects for the Gradient and Cryo Cooler are in the ON position. If not, set to ON as with the TSCC Main Disconnect procedure in step 3, (see Illustration 9 for locations).
5. Place the Control Power Switch into the ON position. TSCC should start after 5 seconds and be fully functional after 30 seconds. If the TSCC shuts down after 30 seconds, refer to section 6-1-4.

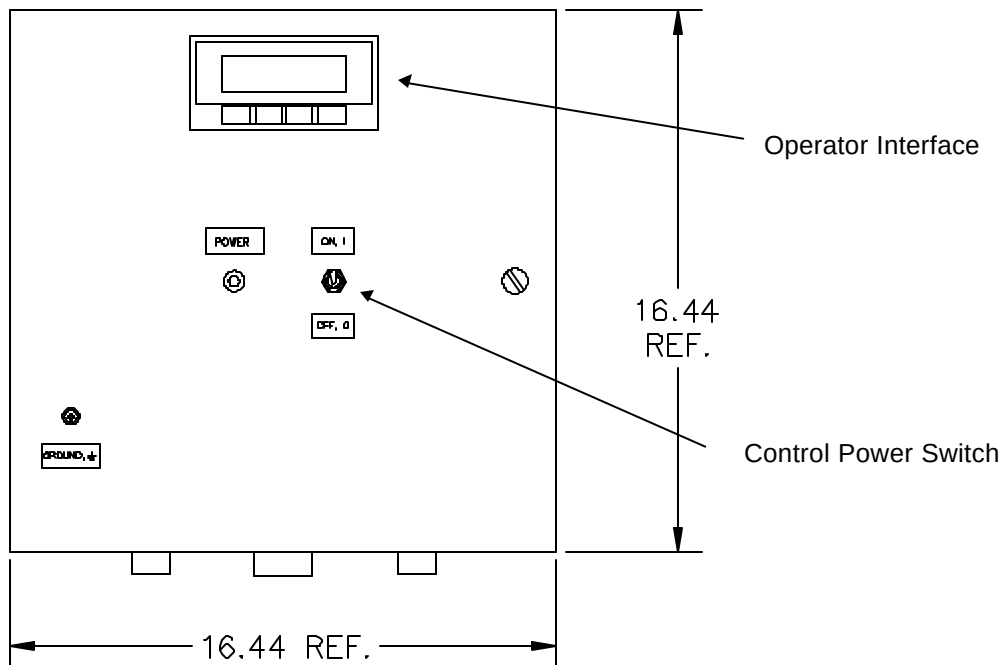


Illustration 13 CONTROL PANEL

6. After the initial 30 seconds of operation, set power switches to ON for the Cryo Cooler.
7. After 30 minutes of operation verify that the unit is maintaining temperatures as in the following table.

NOTE: Two globe valves are located on the coolant supply lines (gradient and cryo loops) for flow adjustment. The valves have been set for 60 Hz operation. For 50 Hz operation, the valves should be adjusted to obtain the flow shown in Table 16.

	Coolant Flow	Coolant Temperature
Gradient unit	6.1 GPM + 0.5 GPM	20° +/- 1C
Cryo Cooler Flow	2.1 GPM + 0.5 GPM	20° +/- 1C

Table 16 **TSCC TEMPERATURE AND FLOW TOLERANCES**

6-1-2 Shutting Down The TSCC

1. Switch to OFF the Gradient and Cryo Coolers.
2. Verify that the red led on the Operator Interface is not on. If the led is glowing, check the alarm section of the Operator Interface for any faults and log them before proceeding. Refer to section 6-1-4 for further instructions.
3. Set the Control Power Switch to OFF.

6-1-3 Maintenance

An approved service provider should conduct routine maintenance every 4 months. This maintenance consists of:

Verifying proper flows and temperatures per table 14 and logging data.

Change coolant filter.

Leak test refrigerant system.

Check electrical control panel for burned or frayed wiring and loose connections.

Checking all coolant hoses for obvious leaks and for worn areas.

Verifying proper operation of refrigeration system.

Check condenser coils for dirt or damage.

Contact GE for current schedule or approved service provider.

6-1-4 General Troubleshooting

The TSCC chiller system is designed to be monitored by the GE Service Center at all times. Some occasions arise where the operator can assist in troubleshooting. In all cases the operator will be most useful by checking for and logging any faults on the Operator Interface and noting any abnormal conditions in the equipment room where the TSCC is located. Refer to sections 6-2-1 through 6-2-3 for instruction on how to locate faults on the Operator Interface screens.

6-2 PLC Operation

6-2-1 Operator Interface Display Description:

The operator interface is located on the front panel of the SCC1 and displays the model number of the unit, the software revision of the programmed PLC, parameter set points, parameter data, parameter alarm status, and system run time. The interface also allows the user to change the operating set points for the fluid coolant circuit. The upper and lower alarm limits for the various parameters are preset by the manufacturer and are not accessible by service. See Illustration 14 for an example of the layout of the display, user buttons, power switch and indicator lights.

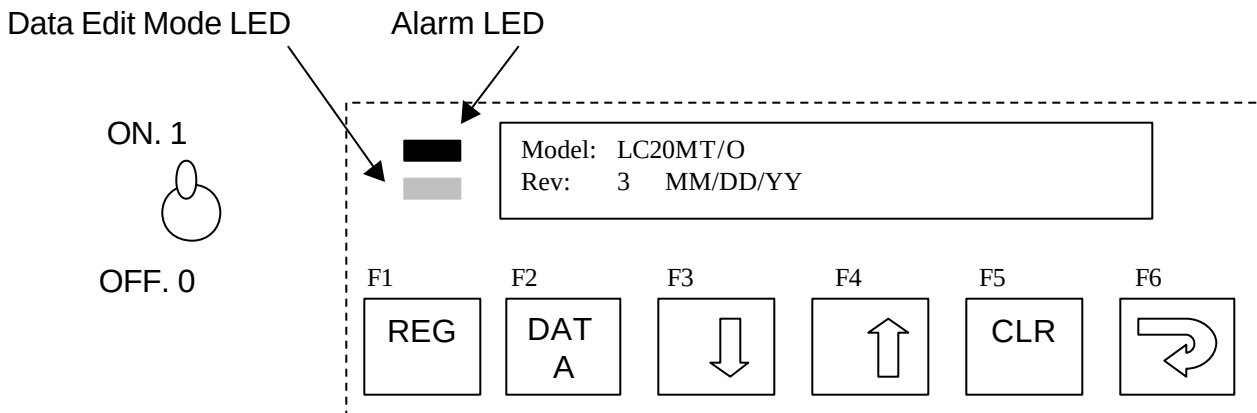


Illustration 14 **OPERATOR INTERFACE**

6-2-2 Button Description

- Button F1 Toggles the display between Menu 1 and Menu 2.
- Button F2 Puts the unit into data edit mode for changing the operating set point for supply coolant temperature. The green led will light when the unit is in data edit mode.
- Button F3 Menu List Display: Steps through the menu list in the forward direction. Change Set Point: Increases the value shown on the display in 0.1 unit increments.
- Button F4 Menu List Display: Steps through the menu list in the reverse direction. Change Set Point: Decreases the value shown on the display in 0.1 unit increments.

- Button F5 To extinguish the red alarm status indicator. **NOTE:** The operating parameter must be in a normal or no fault mode in order to turn off the red alarm status indicator.
- Button F6 Stores new set point changes and exits the data edit mode.

6-2-3 Menu Lists

1. Press the F1 Button to change between Menu 1 and Menu 2.
2. Refer to Table 17, Menu 1 – System Parameter Data, to view the listing of the following types of items: Model Number and Software revision of the unit, operating set points for specific parameters, current parameter data measurement, and total run hours. The first line item on Menu 1 will show the Model Number of the Unit.
3. Refer to Table 18, Menu 2 – System Operating Status, to view the listing of the alarm or failure conditions. The first line item on Menu 2 will show Supply Fluid Temperature.

6-2-4 Changing The Set Points

The coolant temperature set points are preset at Ellis and Watts per G.E. specifications. There is a 2°C allowable variance, settable by the operator. If there is a need to change the set point value, then refer to the following steps to perform the change.

1. Press F3 to step forward through the menu until the display shows the coolant temp set point or the TSCC air temp set point.
2. Press F2 to enter the data edit mode.
3. Press either the F3 or F4 buttons to change the set point value.
4. Press F6 to store the set point values and exit the data edit mode.

Example of Model Number and Revision	
Model Number:	LC20MT/O
Rev.: #	Mo/Dy/Yr
Parameter	Range Value / Unit of Measure (see <u>NOTE 1</u>)
Supply Temp:	0 to 29.4 Degrees Centigrade
Return Temp:	0 to 50 Degrees Centigrade
Temp Set Point:	17 to 24 Degrees Centigrade (Factory set 20°C)
Gradient Flow:	0 to 99.99 GPM
Cryo-Cooler Flow:	0 to 99.99 GPM
TSCC 3 way valve % open:	0 to 100%
Total Run Hours:	0 to 9999 Hours <u>NOTE 2</u>
NOTE 1: Unit of measure is set at the factory.	
NOTE 2: After the run hours reaches 9999, the unit resets to zero.	

Table 17 **MENU 1 – SYSTEM PARAMETER DATA**

Parameter	Display of Operating Model		
Supply Temp:	Low	Normal	High
Coolant Level:	Low	Normal	
Gradient Flow:	Failure	Low	Normal
Cryo-Cooler Flow:	Failure	Low	Normal
Suction Pressure	Fault	No Fault	
Discharge Pressure	Fault	No Fault	
Compressor / Phase	Fault	No Fault	
Compressor Cycling:	Normal	- - - - (See Note 1)	
Note 1: Each dashed line represents an attempt by the PLC to restart the chiller. After 5 attempts the SCC will shut down.			

Table 18 **MENU 2 – SYSTEM OPERATING STATUS**

TERMS & ACRONYMS

GEMS	General Electric Medical Systems
MDP	Main Distribution Panel
RCP	Remote Control Panel
QD	Quick disconnect coupler
RGA	Returned Goods Authorization
EPA	Environmental Protection Agency
MRI	Magnetic Resonance Imaging
UBC	United Builders Code
SST	Stainless Steel
NPTF	National Pipe Thread, Female
DB	Distribution Block
TSCC	Twin System Cooling Cabinet
PLC	Programmable Logic Control
Liquidtite	Trade name for water tight flexible conduit
LocTite	Brand name for adhesives, sealants, and threadlockers

INDEX OF TABLES

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MATERIAL SAFETEY DATA SHEETS

Forane ® R - 134a
Tetrafluoroethane (HFC - 134a)

Dowfrost ® Heat Transfer Fluid
Propylene glycol
Dipotassium Phosphate
Deionized Water

Material Safety Data Sheet

The Dow Chemical Company

Midland, Michigan 48674

Emergency 517 - 636-4400

1. CHEMICAL PRODUCT & COMPANY IDENTIFICATION

Page: 1

24-Hour Emergency Phone Number: 989-636-4400

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99

Date Printed: 05/07/01

MSD: 000130

The Dow Chemical Company, Midland, MI 48674

Customer Information Center: 800-258-2436

2. COMPOSITION/INFORMATION ON INGREDIENTS

Propylene glycol	CAS# 000057-55-6	95%
Dipotassium phosphate	CAS# 007758-11-4	<5%
Deionized water	CAS# 007732-18-5	<5%

3. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW

* Colorless, odorless liquid. Toxic fumes released in fire * situations.

POTENTIAL HEALTH EFFECTS (See Section 11 for toxicological data.)

EYE: May cause slight transient eye irritation. Corneal injury is unlikely.

SKIN: Essentially nonirritating to skin on prolonged contact. A single prolonged skin exposure is not likely to result in the material being absorbed through skin in harmful amounts. Repeated exposures may cause slight flaking, tenderness and softening of skin.

(Continued on page 2)

OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

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low

Product: **DOWFROST* HEAT TRANSFER FLUID** Product Code: **23545**

Effective Date: 01/08/99

Date Printed: 05/07/01

MSD: **000130**

INGESTION: Single dose oral toxicity is low. No hazards anticipated from swallowing small amounts incidental to normal handling operations.

INHALATION: A single prolonged (hours) inhalation exposure is not likely to cause adverse effects. Mists are not likely to be hazardous.

SYSTEMIC (OTHER TARGET ORGAN) EFFECTS: Repeated excessive ingestion may cause central nervous system effects.

CANCER INFORMATION: Did not cause cancer in long-term animal studies.

TERATOLOGY (BIRTH DEFECTS): Birth defects are unlikely. Exposures having no adverse effects on the mother should have no effect on the fetus.

REPRODUCTIVE EFFECTS: In animal studies, has been shown not to interfere with reproduction.

4. FIRST AID

EYE: Flush eyes with plenty of water.

SKIN: Wash off in flowing water or shower.

INGESTION: If swallowed, seek medical attention. Do not induce vomiting unless directed to do so by medical personnel.

INHALATION: Remove to fresh air if effects occur. Consult a physician.

NOTE TO PHYSICIAN: No specific antidote. Supportive care. Treatment based on judgment of the physician in response to reactions of the patient.

5. FIRE FIGHTING MEASURES

FLASH POINT: 215F, 102C METHOD USED: TOC (Tag Open Cup)

(Continued on page 3)

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FLAMMABLE LIMITS LFL: 2.6% @ 100C UFL: 12.5% @ 130C

HAZARDOUS COMBUSTION PRODUCTS: During a fire, smoke may contain the original material in addition to unidentified toxic and/or irritating compounds. Hazardous combustion products may include and are not limited to carbon monoxide, carbon dioxide.

OTHER FLAMMABILITY INFORMATION: Violent steam generation or eruption may occur upon application of direct water stream. Flammable concentration of vapor can accumulate at temperatures above 215.0 deg. F. Liquid mist of this product can burn. Spills of these organic liquids on hot fibrous insulations may lead to lowering of the autoignition temperatures possibly resulting in spontaneous combustion.

EXTINGUISHING MEDIA: Water fog or fine spray, carbon dioxide, dry chemical, foam. Alcohol resistant foams (ATC type) are preferred if available. General purpose synthetic foams (including AFFF) or protein foams may function, but much less effectively. Do not use direct water stream. Will spread fire.

MEDIA TO BE AVOIDED: Do not use direct water stream.

FIRE FIGHTING INSTRUCTIONS: Keep people away. Isolate fire area and deny unnecessary entry. Burning liquids may be moved by flushing with water to protect personnel and minimize property damage. Burning liquids may be extinguished by dilution with water. Do not use direct water stream. May spread fire.

PROTECTIVE EQUIPMENT FOR FIRE FIGHTERS: Wear positive-pressure self-contained breathing apparatus (SCBA) and protective firefighting clothing (includes fire-fighting helmet, coat, pants, boots, and gloves). If protective equipment is not available or not used, fight fire from a protected location or safe distance.

6. ACCIDENTAL RELEASE MEASURES (See Section 15 for Regulatory Information)

(Continued on page 4)

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PROTECT PEOPLE: Clear non-emergency personnel from area.

PROTECT THE ENVIRONMENT: Contain liquid to prevent contamination of soil, surface water or ground water.

CLEANUP: Clean up with absorbent material. Sweep up.

7. HANDLING AND STORAGE

HANDLING: Product on surfaces can cause slippery conditions.

STORAGE: Store below 121 C, 250 F.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

ENGINEERING CONTROLS: Good general ventilation should be sufficient for most conditions.

PERSONAL PROTECTIVE EQUIPMENT

EYE/FACE PROTECTION: Use safety glasses.

SKIN PROTECTION: For brief contact, no precautions other than clean body-covering clothing should be needed. Use impervious gloves when prolonged or frequently repeated contact could occur.

RESPIRATORY PROTECTION: No respiratory protection should be needed.

EXPOSURE GUIDELINE(S): Propylene glycol: AIHA WEEL is 50 ppm total, 10 mg/m3 aerosol only.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE: Colorless. ODOR: Odorless liquid. VAPOR PRESSURE: 0.22 mmHg @ 20C, 68F
VAPOR DENSITY: 2.62 BOILING POINT: 370F, 188C SOLUBILITY IN WATER: Complete

(Continued on page 5)

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SPECIFIC GRAVITY: 1.050 @ 60/60F, 16C

10. STABILITY AND REACTIVITY

CHEMICAL STABILITY: Thermally stable at typical use temperatures.

CONDITIONS TO AVOID: Avoid temperatures above 121C/250F. Product can decompose at elevated temperatures.

INCOMPATIBILITY WITH OTHER MATERIALS: Avoid contact with oxidizing materials.

HAZARDOUS DECOMPOSITION PRODUCTS: Hazardous decomposition products depend upon temperature, air supply and the presence of other materials.

HAZARDOUS POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL INFORMATION (See Section 3 for Potential Health Effects. For detailed toxicological data, write or call the address or non-emergency number shown in Section 1)

SKIN: The LD50 for skin absorption in rabbits is greater than 10,000 mg/kg.

INGESTION: The oral LD50 for female rats is about 20.3 g/kg.

MUTAGENICITY (EFFECTS ON GENETIC MATERIAL): Results of in vitro (test tube) mutagenicity tests have been negative. Results of mutagenicity tests in animals have been negative.

12. ECOLOGICAL INFORMATION (For detailed Ecological data, write or call the address or non-emergency number shown in Section 1)

ENVIRONMENTAL FATE MOVEMENT AND PARTITIONING: Based largely or completely on data for major component(s). Bioconcentration potential is low (BCF less than 100 or Log Pow less than 3).

DEGRADATION AND PERSISTENCE: Based largely or completely on data for major component(s). Biodegradation under aerobic static laboratory conditions is high (BOD20 or BOD28/ThOD

(Continued on page 6)

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greater than 40%).

ECOTOXICITY: Based largely or completely on data for major component(s). Material is practically non-toxic to aquatic organisms on an acute basis (LC50 greater than 100 mg/L in most sensitive species).

13. DISPOSAL CONSIDERATIONS (See Section 15 for Regulatory Information)

DISPOSAL: DO NOT DUMP INTO ANY SEWERS, ON THE GROUND OR INTO ANY BODY OF WATER. All disposal methods must be in compliance with all Federal, State/Provincial and local laws and regulations. Regulations may vary in different locations. Waste characterizations and compliance with applicable laws are the responsibility solely of the waste generator. THE DOW CHEMICAL COMPANY HAS NO CONTROL OVER THE MANAGEMENT PRACTICES OR MANUFACTURING PROCESSES OF PARTIES HANDLING OR USING THIS MATERIAL. THE INFORMATION PRESENTED HERE PERTAINS ONLY TO THE PRODUCT AS SHIPPED IN ITS INTENDED CONDITION AS DESCRIBED IN MSDS SECTION 2 (Composition/information On Ingredients).

FOR UNUSED & UNCONTAMINATED PRODUCT, the preferred options include sending to a licensed, permitted: recycler, reclaimer, or waste water treatment system.

As a service to its customers, Dow can provide names of information resources to help identify waste management companies and other facilities which recycle, reprocess or manage chemicals or plastics, and that manage used drums. Telephone Dow's Customer Information Center at 800-258-2436 or 517-832-1556 for further details.

14. TRANSPORT INFORMATION

CANADIAN TDG INFORMATION: For TDG regulatory information, if required, consult transportation regulations, product shipping papers, or your Dow representative.

15. REGULATORY INFORMATION (Not meant to be all-inclusive- -selected regulations represented)

NOTICE: The information herein is presented in good faith and

(Continued on page 7)

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Product: DOWFROST* HEAT TRANSFER FLUID

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believed to be accurate as of the effective date shown above. However, no warranty, express or implied is given. Regulatory requirements are subject to change and may differ from one location to another; it is the buyer's responsibility to ensure that its activities comply with federal, state or provincial, and local laws. The following specific information is made for the purpose of complying with numerous federal, state or provincial, and local laws and regulations. See other sections for health and safety information.

U.S. REGULATIONS

SARA 313 INFORMATION: To the best of our knowledge, this product contains no chemical subject to SARA Title III Section 313 supplier notification requirements.

SARA HAZARD CATEGORY: This product has been reviewed according to the

EPA "Hazard Categories" promulgated under Sections 311 and 312 of the Superfund Amendment and Reauthorization Act of **1986** (SARA Title 111) and is considered, under applicable definitions, to meet the following categories:
Not to have met any hazard category

TOXIC SUBSTANCES CONTROL ACT (TSCA):
All ingredients are on the TSCA inventory or are not required to be listed on the TSCA inventory.

STATE RIGHT-TO-KNOW: The following product components are cited on certain state lists as mentioned. Non-listed components may be shown in the composition section of the MSDS.

(Continued on page 8)

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Product: DOWFROST* HEAT TRANSFER FLUID

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REGULATORY INFORMATION (CONTINUED)

CHEMICAL NAME

CAS NUMBER LIST

1,2-PROPANEDIOL

000057-55-6 PA1

PA1=Pennsylvania Hazardous Substance (present at greater than or equal to 1.0%).

CANADIAN REGULATIONS

WHMIS INFORMATION: The Canadian Workplace Hazardous Materials Information System (WHMIS) Classification for this product is:

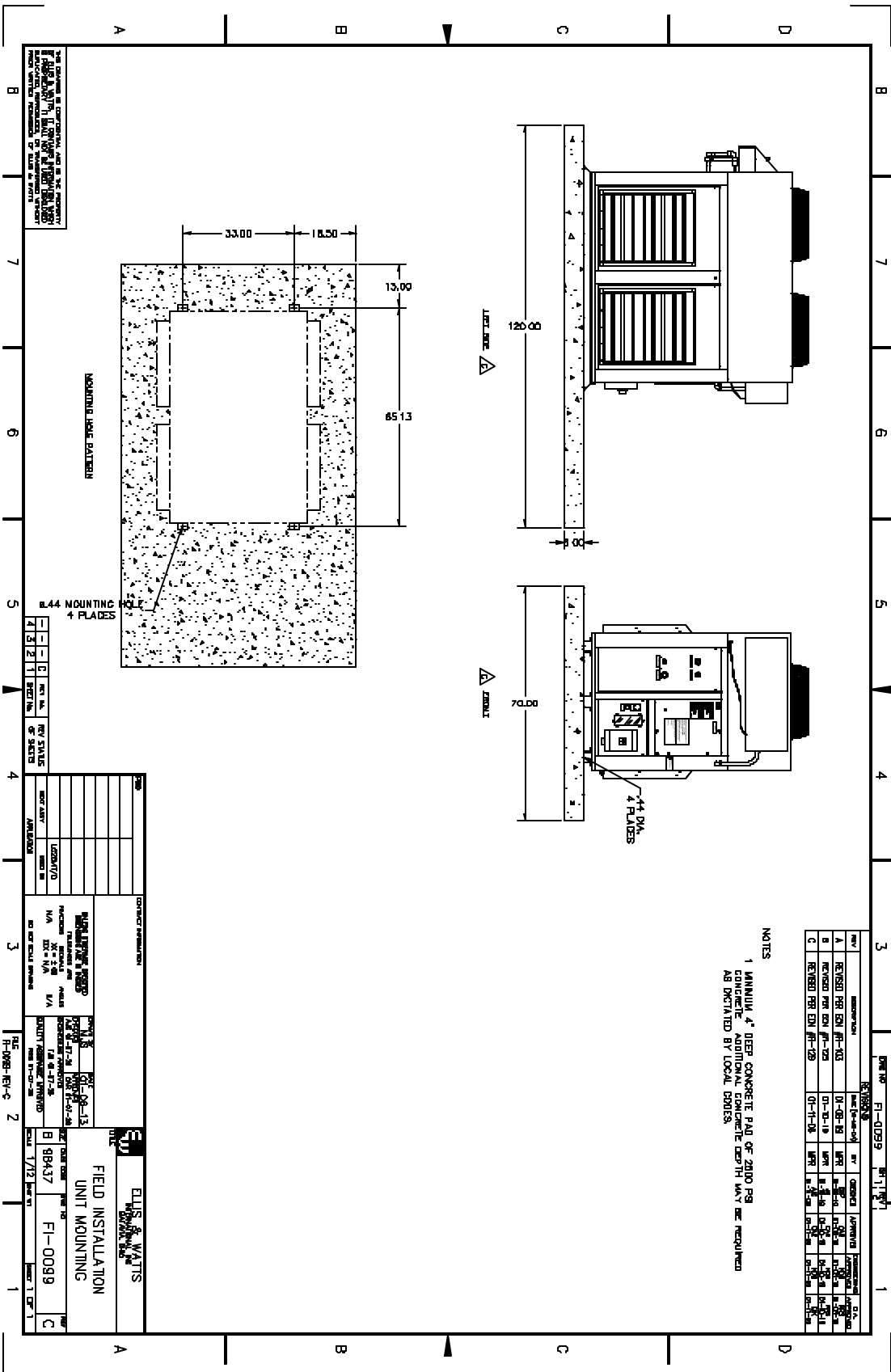
This product is not a "Controlled Product" under WHMIS.

16. OTHER INFORMATION

MSDS STATUS: Revised Section 13.

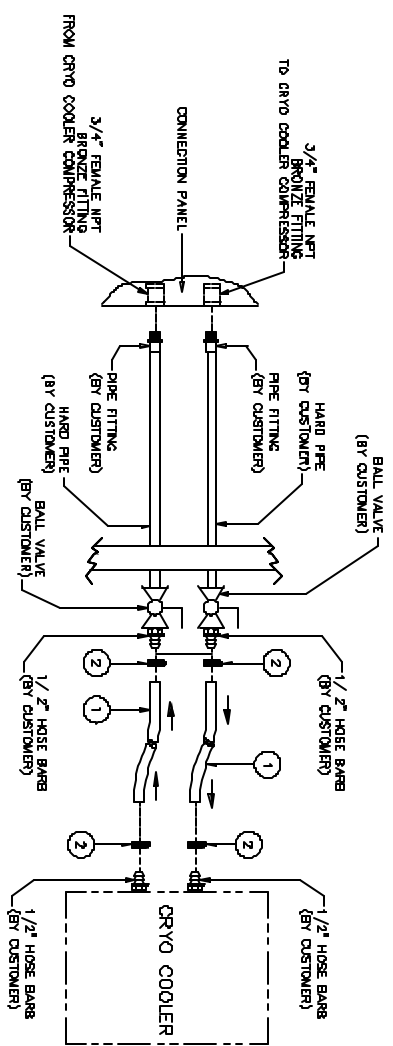
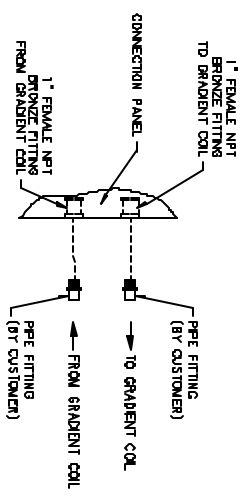
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FI DRAWINGS



REV	DESCRIPTION	DATE	BY	CHKD	APPROVED	REVISION
1	REVISED PER EDN F-121	04-10-21	WPR	WPR	WPR	1
2	REVISED PER EDN F-131	01-11-22	WPR	WPR	WPR	2

- NOTES
- USE LOGITE 587 HIGH TEMPERATURE THREAD SEALANT OR EQUAL ON ALL THREADED FITTINGS.
 - FOR Piping SCHEMATIC REFERENCE DRAWING J1538-1301 OR J1538-1301



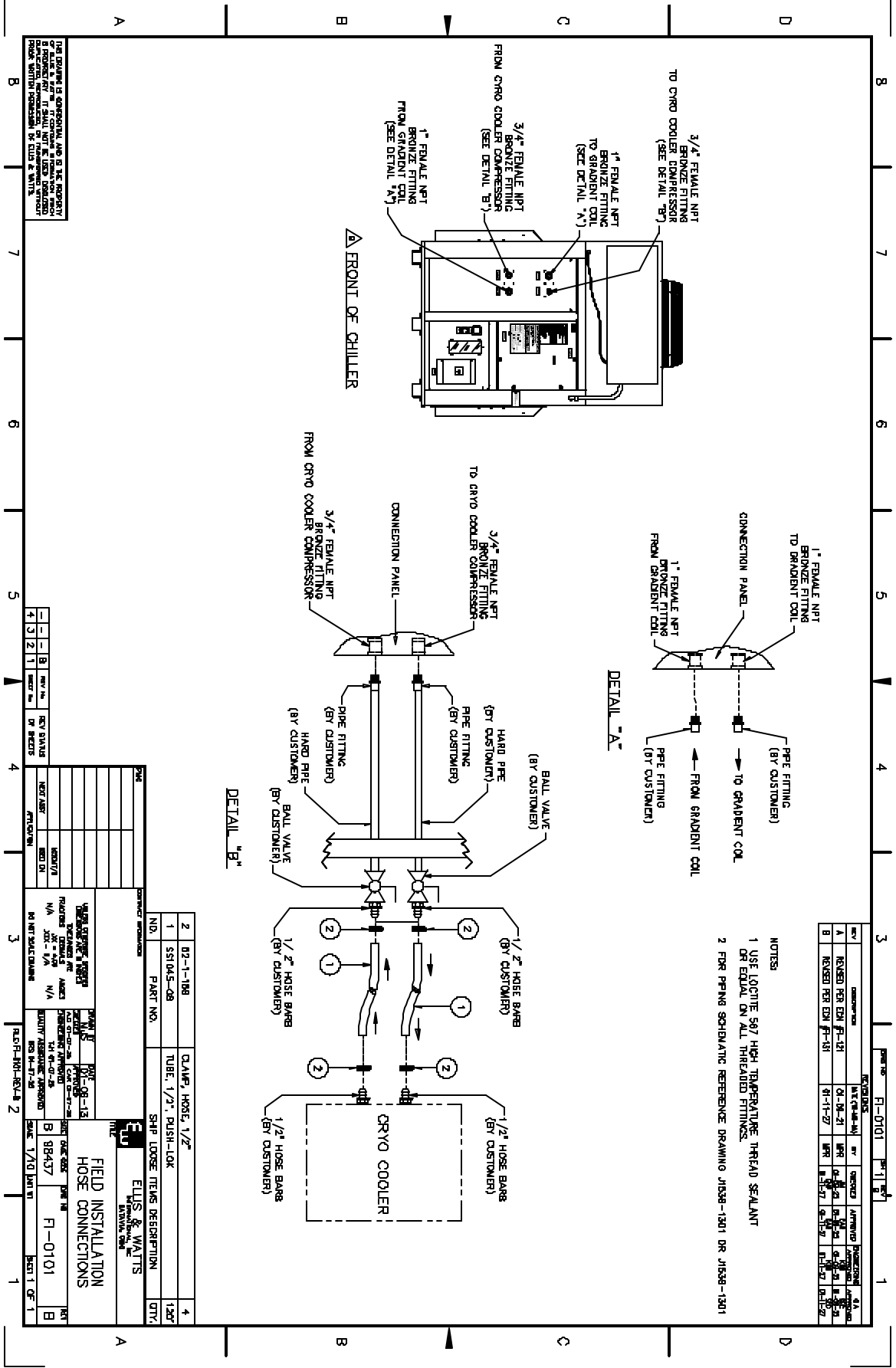
Δ FRONT OF CHILLER

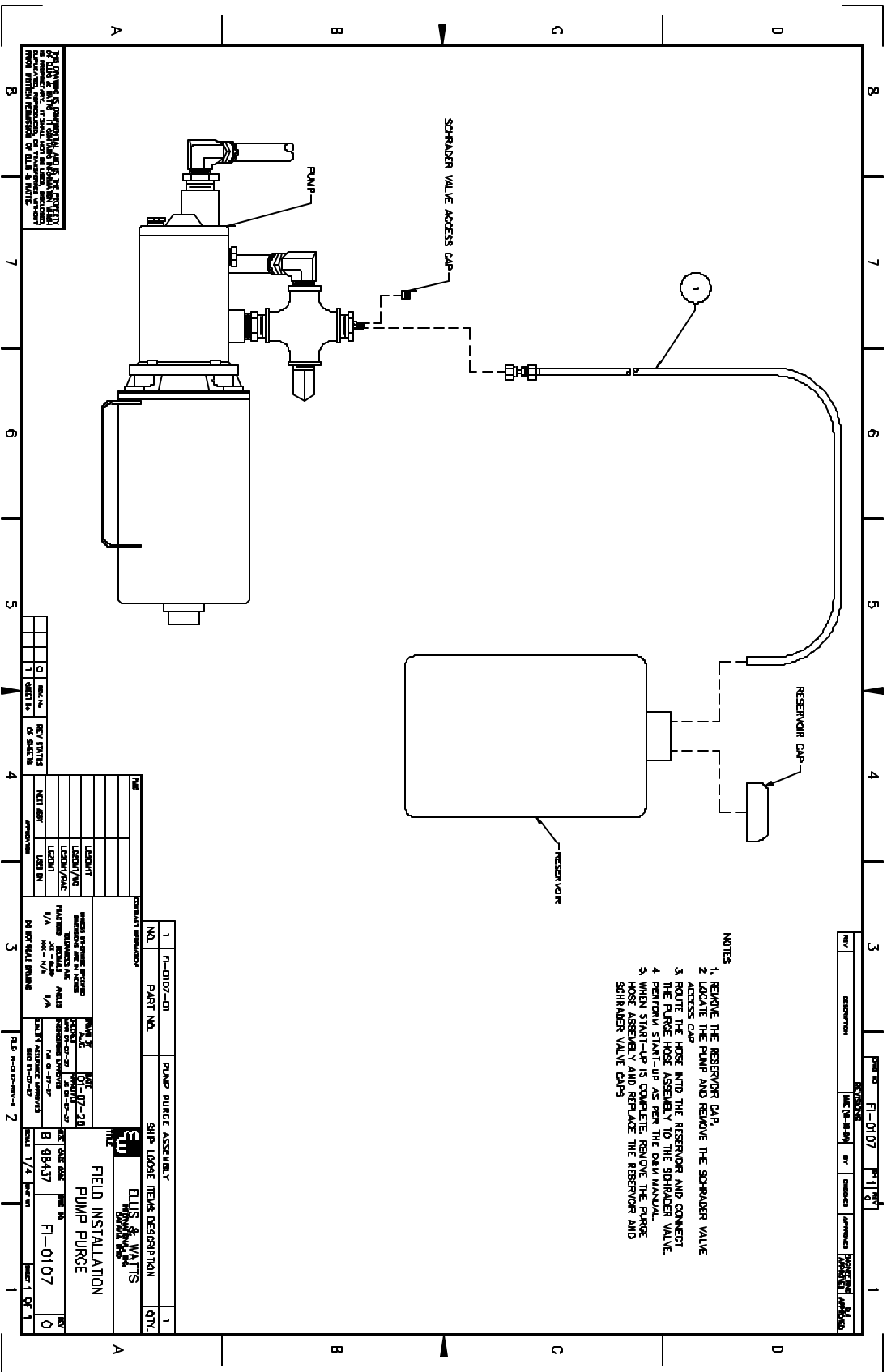
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REV	DESCRIPTION	DATE	BY	CHKD	APPROVED	REVISION
1	REVISED PER EDN F-121	04-10-21	WPR	WPR	WPR	1
2	REVISED PER EDN F-131	01-11-22	WPR	WPR	WPR	2

2	02-1-1-108	CLAMP, HOSE, 1/2"				4
1	S51045-08	TUBE, 1/2", PUSH-LOCK				1207
	PART NO.	SHIP LOGGE TENS DESCRIPTION				QTY

1	02-1-1-108	CLAMP, HOSE, 1/2"				4
1	S51045-08	TUBE, 1/2", PUSH-LOCK				1207
	PART NO.	SHIP LOGGE TENS DESCRIPTION				QTY





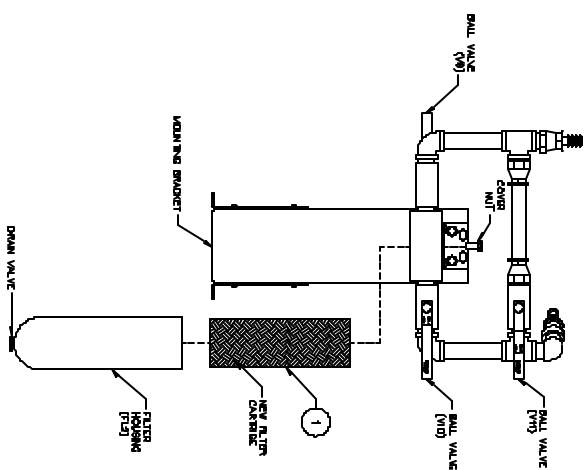
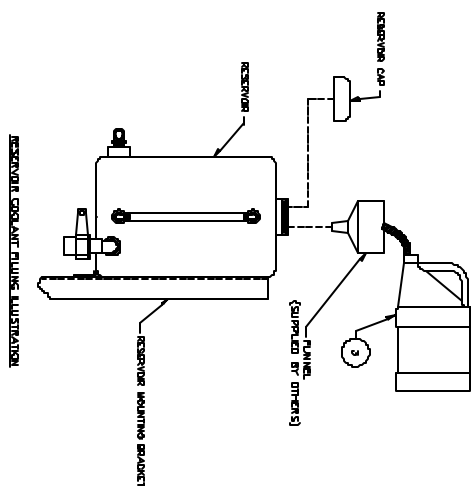
- NOTES
1. REMOVE THE RESERVOIR CAP.
 2. LOCATE THE PUMP AND REMOVE THE SCHRAEDER VALVE ACCESS CAP
 3. ROUTE THE HOSE INTO THE RESERVOIR AND CONNECT THE PUMP HOSE ASSEMBLY TO THE SCHRAEDER VALVE.
 4. PERFORM START-UP AS PER THE DRAIN MANUAL.
 5. WHEN START-UP IS COMPLETE REMOVE THE PUMP HOSE ASSEMBLY AND REPLACE THE RESERVOIR AND SCHRAEDER VALVE CAPS

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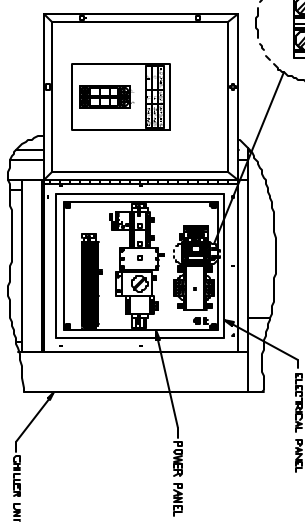
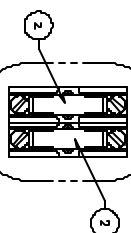
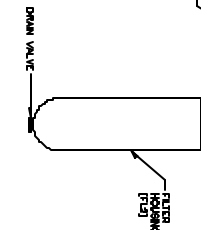
REV	DATE	BY	CHKD	APP'D
1	01-07-20	W		

REV	DATE	BY	CHKD	APP'D
1	01-07-20	W		

1	FL-0107-01	PUMP PURGE ASSEMBLY	QTY. 1
NOL	PART NO.	SHIP LOOSE ITEMS DESCRIPTION	
Elsa & Waits			
FIELD INSTALLATION			
PUMP PURGE			
REV 1 OF 1			



- NOTES
1. LOCATING FILTER REPLACEMENT:
 - A. REMOVE SIDE PANEL OF THE UNIT TO ACCESS FILTER AND PUMP.
 - B. CONNECT PUMP PUMP ASSEMBLY AS PER R-0107
 - C. OPEN VALVE WITH LOCATED ABOVE FILTER.
 - D. PUMP MUST BE OPERATING FOR 10 MINUTES OF THE FILTER.
 - E. PUMP MUST BE UNDER THE FILTER.
 - F. REMOVE FILTER HOUSING. MUST CHANGE THE FILTER HOUSING
 - G. PUMP MUST BE OPERATING.
 - H. PUMP MUST BE UNDER THE DISCARD AS PER THE PROBLEMING
 - I. PUMP MUST BE UNDER THE DISCARD.
 - J. PUMP MUST BE UNDER THE DISCARD.
 - K. PUMP MUST BE UNDER THE DISCARD.
 - L. PUMP MUST BE UNDER THE DISCARD.
 - M. PUMP MUST BE UNDER THE DISCARD.
 - N. PUMP MUST BE UNDER THE DISCARD.
 - O. PUMP MUST BE UNDER THE DISCARD.
 - P. PUMP MUST BE UNDER THE DISCARD.
 - Q. PUMP MUST BE UNDER THE DISCARD.
 - R. PUMP MUST BE UNDER THE DISCARD.
 - S. PUMP MUST BE UNDER THE DISCARD.
 - T. PUMP MUST BE UNDER THE DISCARD.
 - U. PUMP MUST BE UNDER THE DISCARD.
 - V. PUMP MUST BE UNDER THE DISCARD.
 - W. PUMP MUST BE UNDER THE DISCARD.
 - X. PUMP MUST BE UNDER THE DISCARD.
 - Y. PUMP MUST BE UNDER THE DISCARD.
 - Z. PUMP MUST BE UNDER THE DISCARD.
2. FILTER REPLACEMENT:
 - A. DISCONNECT AND LOCATING ABOUT REMOVED POWER MAT
 - B. REMOVE FILTER HOUSING.
 - C. OPEN POWER PANEL, ACCESS PANEL.
 - D. REMOVE PUMP WITH A PUMP OF THE PUMP.
 - E. PUMP MUST BE UNDER THE DISCARD.
 - F. PUMP MUST BE UNDER THE DISCARD.
 - G. PUMP MUST BE UNDER THE DISCARD.
 - H. PUMP MUST BE UNDER THE DISCARD.
 - I. PUMP MUST BE UNDER THE DISCARD.
 - J. PUMP MUST BE UNDER THE DISCARD.
 - K. PUMP MUST BE UNDER THE DISCARD.
 - L. PUMP MUST BE UNDER THE DISCARD.
 - M. PUMP MUST BE UNDER THE DISCARD.
 - N. PUMP MUST BE UNDER THE DISCARD.
 - O. PUMP MUST BE UNDER THE DISCARD.
 - P. PUMP MUST BE UNDER THE DISCARD.
 - Q. PUMP MUST BE UNDER THE DISCARD.
 - R. PUMP MUST BE UNDER THE DISCARD.
 - S. PUMP MUST BE UNDER THE DISCARD.
 - T. PUMP MUST BE UNDER THE DISCARD.
 - U. PUMP MUST BE UNDER THE DISCARD.
 - V. PUMP MUST BE UNDER THE DISCARD.
 - W. PUMP MUST BE UNDER THE DISCARD.
 - X. PUMP MUST BE UNDER THE DISCARD.
 - Y. PUMP MUST BE UNDER THE DISCARD.
 - Z. PUMP MUST BE UNDER THE DISCARD.
3. RECONNECT LOCATING FILTER:
 - A. REMOVE RIGHT SIDE ACCESS PANEL. (TWO 1/4 TURN PARTS).
 - B. REMOVE THE DISCARD MAT.
 - C. PLACE FILTER IN THE RECONNECT FILTER. (PUMPED BY OTHERS)
 - D. PUMP MUST BE UNDER THE DISCARD.
 - E. PUMP MUST BE UNDER THE DISCARD.
 - F. PUMP MUST BE UNDER THE DISCARD.
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 - Z. PUMP MUST BE UNDER THE DISCARD.



EUR-REPLACEMENT ILLUSTRATION

REV	STATUS	BY	DATE
4	3	2	1

PANEL	
NEXT ASSY	L2020M7/D L2020 BN
©FACOM-IBM	

[illegible]

ELIS & WATTS		ELIS & WATTS	
WINDHAM, VT		WINDHAM, VT	
DANIEL PERD		DANIEL PERD	
FIELD INSTALLATION		FIELD INSTALLATION	
SPARE PARTS KIT		SPARE PARTS KIT	
QTY	UNIT PRICE	QTY	UNIT PRICE
8	984.37	0	984.37
F1-0112		F1-0112	
TOTAL 1 OF 1		TOTAL 1 OF 1	

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